Blog Snippets Based Drug Effects Extraction System Using Lexical and Grammatical Restrictions

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ABSTRACT

Obtaining medical information has a beneficial influence on patients' treatment and QOL (quality of life). The authors aim to make a system that helps patients to collect narrative information. Extracting information from data written by patients will allow the acquisition of information which is easy to understand and provides encouragement. Additionally, by using large-scale data, the system can be utilized for discovering unknown effects or patterns. As the first step, the purpose of this paper is to extract descriptions of the effects caused by taking drugs as a triplet of expressions from illness survival blogs' snippets. This paper proposes a method to extract the triplets using specific clue words and parsing the results in order to extract from blogs written in free natural language. Moreover, recall was improved by combining the their proposed method and a baseline system, and precision was improved by filtering using dictionaries we created from existing medical documents.

Keywords: *Blog Snippets, Drug Effects, Information Extraction, Japanese, Medical Information, Opinion Mining, Text Mining, Triplet Extraction*

1. INTRODUCTION

Currently, the total number of patients with serious illnesses in Japan has reached about 18,026,000 (Ministry of Health, Labour and Welfare, 2011). When someone is diagnosed with a disease or feels that they might be ill, medical information is required to solve various questions in such situations. If patients obtain the required information and they are satisfied with it, this can motivate patients in decision-making and proactive treatment, and have beneficial effects for their QOL (quality of life) and health (Setoyama & Nakayama, 2011). Most major sources of medical information for patients are medical personnel, who are

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recognized as a reliable source (Taniguchi, 2004; Hesse et al., 2005). However, in order to make better decisions, both "evidence information", which shows the effects for the population stochastically, and "narrative information" which consists of individual stories from personal experience are necessary (O'Connor, 2002). Although obtaining stories of the experiences of other patients with the same illness can reduce a patient's anxiety and encourage them (Matsumoto et al., 2005; Maeda et al., 2009), it is difficult for medical staff to provide patients with enough narrative information. Therefore, patients are using various sources such as the Internet in addition to doctors, as has been clarified by research (Setoyama & Nakayama, 2011; Walsh et al., 2010). However, it is not easy to extract and collect answers required by patients appropriately, since there is a huge amount of information on the Web, and it is written in free natural language. In addition, we can see from survey results that it is hard for patients to ask doctors or nurses certain questions (Tsuchiya & Horn, 2009; Kokubu 2008), and we can also see that patients feel that there is a lack of information. Accordingly, we aim to make a system that helps patients to obtain information about other patients with the same illness.

Currently, pharmaceuticals used in Japan consist of about 18,000 prescription drugs, which are prescribed based on a diagnosis of a doctor, and about 12,000 OTC (Over-the-Counter) drugs, which people can buy at pharmacies and drugstores without a prescription by a doctor. Although these drugs are necessary for medical treatment, side effects that occur with use are a problem. Early detection of side effects and taking countermeasures are important issues in the medical field (Mikami et al., 2013). Thus, a great deal of research to estimate the relationship between drugs and side effects has been carried out, and there are many medicine search services (NPO Narrative of Disease and Health DIPEx-Japan, n.d.; Pharmaceuticals and Medical Devices Agency, n.d.). These are based on data created by manufacturers and distributors of pharmaceuticals and information reported by doctors or pharmacists. As yet, medical institutions cannot deal with unknown side effects because additional risks of using medication, such as those not reported, often become known after a product is used by a larger number of patients for a longer period of time. Therefore, we need to collect as much information as possible in order to ascertain effects and side effects of medicines clearly. Thus, we focused on large-scale data on the Web which we can obtain in real time. We extract medical information from blogs written by patients and their families in order to acquire narrative information. Although patients' understanding of diseases and medical care and the reliability of Internet sources are still far from satisfactory, information on the Web is extensive and instantaneous, and thus can be used to understand people's changes and trends. Additionally, Japanese people usually disclose personal information anonymously on social media and blogs. It is considered that anonymous narratives are unreliable due to lacking a sense of responsibility. However, anonymized information on the Web gives us information that may be inaccessible in the real world, since anonymity allows users to express opinions and ideas that they may not want someone close to know, without the need to worry about appearances (Orita et al., 2007).

We aim to develop a system that searches for medical information from patients' blogs. For example, in our proposed system, when the user inputs a name of a disease, the system collects blogs written by patients of that disease, extracts information related to treatment from them, and outputs statistical data such as a distribution of treatments. Moreover, when the user inputs a drug name, we aim to output data on how many patients felt what kind of effects, side effects, and unknown effects. We believe that a system that searches for drug effects will be very useful for patients' mental support and decision-making regarding their treatment, and would be useful for investigating unknown effects.

As a first step, we intend to extract descriptions of both positive and negative effects caused by taking drugs. The core of an opinion

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typically consists of three elements: an evaluated subject, focused attribute and value (Kobayashi et al., 2004; Sugiki et al., 2008; Tsuchida et al., 2009). For this reason, we extract the effects and changes caused by taking drugs as a triplet of expressions – drug name, object of change, and its effect – from illness survival blogs. Drug effects are sometimes expressed by single words (e.g. "effective", "good", "bad"). However, the polarity sometimes can be read by sets of object of change and relevant effect (e.g. "pain/sensation – loss", "hair – loss"). We extract drug effects expressed by the sets of object phrase and effect phrase ("value" in opinion mining) in order to extract effects which cannot be extracted by simple pattern matching.

2. RELATED WORK

2.1. Relation Inference Systems for Drugs and Side Effects

Unknown side effects are deduced from findings that doctors and pharmacists report by signal detection (Green & Swets, 1966). Signal detection is a statistical method to detect the possibility of a relationship between a medicine and an unproved side effect. However, there is a problem in that it depends on case reports, which can change in number and be biased by social influences. In addition, it is impossible in practice to obtain all data such as those which are not reported. Accordingly, we aim to collect information from blogs written by patients and their families in order to get more medical information. Information on the Internet sent by patients enables us to acquire information that is fresh, updated in real time, and cannot be fully grasped by doctors. This paper presents an approach which extracts drug effects that are considered to be relevant to the drugs by using specific expressions. In future work, we aim to develop a system that estimates the relationship between medicine and extracted effects from sentences which do not include drug names by utilizing already extracted targets and evaluations as a knowledge source.

2.2. Medical Search Systems

There is a website called DIPEx-Japan, which supplies narrative information related to dementia, breast cancer, and prostate cancer. About 150 patients relate their own experiences, showing their faces online (NPO Narrative of Disease and Health DIPEx-Japan, n.d). However, not many people are willing to do this, because Japanese people tend to disclose personal information anonymously. On the other hand, we can acquire information that is not easy to obtain directly, as there is a large amount of anonymized information on weblogs. Moreover, we can cover the low reliability of anonymized information by collecting many related information from the Web.

There are many medical search services provided by companies or corporations which allow the user to search for efficacy and side effects of drugs. One of these is a service for searching the contents of medicine package inserts (Pharmaceuticals and Medical Devices Agency, 2013). Medicine package inserts are documents provided by manufacturers and distributors of pharmaceuticals that contain information about medicines, their ingredients, usage, side effects, and so on. Such medical search systems utilize known data, and as a result, cannot search for unknown effects which may be related. Our system aims to collect information which is useful for detecting unknown effects and side effects via extraction from blogs written by patients.

2.3. Data Mining Systems

When medical search systems or relation inference systems for drugs and side effects are constructed, the specification of the part describing medicinal effects requires considerable efforts. Thus, in recent years, research is being conducted on the medical applications of information technology, with a focus on natural language processing techniques (Kaiser & Bodendorf, 2012; Aramaki et al., 2011; Kin et al., 2013). Aramaki et al. (2010) developed a system that examines adverse effects and

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event information buried in Electronic Health Records (EHR) in hospitals. The relationship between drugs and side effects is usually extracted by support vector machine (SVM) (Cortes & Bapnik, 1995). Shinohara et al. (2011) proposed a method that extracts side effects of drugs using syntactic patterns. In electronic medical records, expressions are recorded in written (formal) language. In blog articles, on the other hand, expressions are often written in spoken (casual) language and non-technical terms. We believe that rare information can be collected by extracting medicinal effects and side effects from texts written not from the doctor's point of view, but from the viewpoint of patients. Because new information that is not documented in drug descriptions can be mined from blog reports and can provide valuable input to determine unknown drug effects.

We can regard drug information obtained by users as evaluation of medicines. Extraction of opinions about certain objects as triplets has been studied. Sugiki et al. (2008) proposed a method for retrieving products that match users' search queries written in natural language using representations that transformed texts of product reviews or queries into triplets ("object", "item", "value"). Tsuchida et al. (2009) proposed a ranking method for opinion retrieval from weblog articles that uses a confidence model of opinion as a three-tuple of object-attribute-evaluation. The method extracts attribute-evaluation pairs (finding evaluation words using their evaluation-expression dictionary which covers about 13,000 words) by SVM (Cortes & Bapnik, 1995), and associates objects with attribute-evaluation pairs by the distance of those words. In studies to extract opinions about a product or location, it is possible to extract using an evaluation-dictionary, since the object ("object of change" in our study) is clear and expressions of evaluation used in texts are limited. Blogs are written not to evaluate the effects of drugs, but to record daily life. Expressions of evaluation that are used to write about drugs or conditions contain words that are used in product reviews, as well as mimetic words. There are no evaluation-dictionaries that include Japanese mimetic words, for example "panpan", meaning "bulging". Moreover, it is not easy to create an evaluation-dictionary that is suitable for medical evaluation, because of the variety of mimetic words.

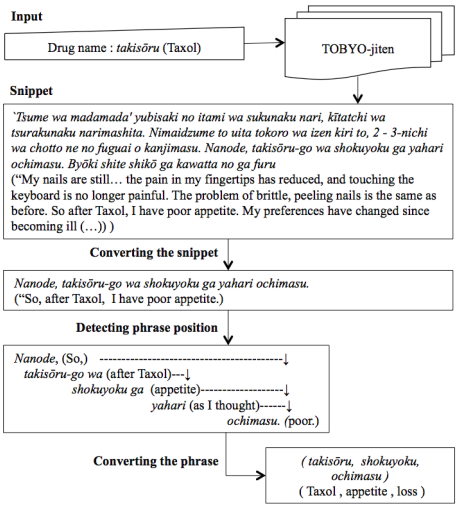
Our proposed system may solve three existing problems in related researches. Firstly, our system can handle texts written in Japanese casual language. Moreover, it covers Japanese onomatopoeia which is an important cue for extracting effects of drug usage and popularly used in blogs. Secondly, unlike the work of Aramaki et al. (2010) and Shinohara et al. (2011) that focused only on negative effects of the drug usage, our system extracts both positive and negative effects. Thirdly, our system is capable of extracting the effects of drug usage even from blogs that include many kinds of topics in one article.

3. PROPOSED METHOD

This paper proposes an approach which uses clue words to enable the extraction of the effects and changes caused by taking drugs as triplets ("drug", "object", "effect") from illness survival blogs. Here, "Object" is a body region or disposition in which the change or effect occurred, and "Effect" is an expression or mental attitude that shows the effect and change of the patient's state. It is not easy to extract useful information automatically and appropriately since blogs are written in different ways, and blog sites have various formats. Accordingly, we extracted from snippets, which are fragments of blogs containing such a query. We collected snippets using TOBYO-jiten, which is a tool for searching for blogs in the TOBYO service. TOBYO is a website on which more than 40,000 patients or family members have registered their blogs about their fight with illness (INITIATIVE INC, 2013). Moreover, since blogs collected by TOBYO-jiten were written by patients and their families, they are useful for developing our system because of noiseless data. Our system searches using any of the 296 drug names registered in TOBYO, and extracts

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Figure 1. Our system overview and example



the blog snippets from the search results using the drug name as a keyword. The sites registered in TOBYO are already tagged with the disease name, patient's gender and the age of the patient when the symptoms first manifested. We plan to use the information from these tags in the next version of our system, which will use a more personalized search.

The extraction system consists of three steps:

1. **Converting the Snippet:** The system converts the snippet to a form that is easy to parse.
2. **Detecting Phrase Position:** The system detects the position of the phrase that includes the word which is an element of the triplet relating to drug effects, using extraction rules that consider the pattern of syntactic dependency and the clue words we set uniquely.
3. **Converting the Phrase:** The system converts the phrase into an appropriate form as an element of the triplet relating to drug effects.

Figure 1 shows an overview and an example of our system. Hereafter, Japanese words will be written in italics. A detailed explanation of the extraction process is discussed later in this section.

In the preliminary research (Kitajima et al., 2012-2013), the first author tagged triplets with 1,000 snippets which were randomly collected using "TOBYO-jiten". As a result, we found that 78.3% of the information on effects of drugs was described in the same sentence that contains the name of the drug. Therefore, we use sentences containing drug name to extract effects.

3.1. Clue Words

In the preliminary research, we found that the drug effects are written after specific nouns and verbs such as "side effects", "after" and "infection" (Kitajima et al., 2012-2013). These imply that sentence contains information about using drugs. A selection of sentences involving the triplet ("drug", "object", "effect") of the information on drug effects is shown in Table 1. We discovered that 76.2% of 181 sentences that contain triplets also contain such specific words (underlined in Table 1). Accordingly, we collected the expressions, and set 32 nouns and verbs (in the dictionary form) as clue words in order to identify the position of the word that contains information on drug effects. Table 2 shows a selection of clue words.

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3.2. Converting the Snippets

Our system converts the snippets which are collected using "TOBYO-jiten" in four different steps in order to reduce parsing mistakes. The first step is using "." as a delimiter. The second step is to identify sentences that contain drug names, and our system extracts triplets from these sentences. In the third step, our system changes the drug name into a placeholder in order to avoid parsing errors caused by drug names that are not registered in the parser dictionary. The last step is to delete parentheses that do not contain drug names (e.g. emoticons such as "(^_^)"). A necessary future task is to conduct assessment of whether deletion will be a problem or not before deleting parentheses, since parentheses may include important information even if they do not contain drug names.

3.3. Detecting Phrase Position

Our system detects dependency chunks that include triplet elements by extraction rules (See Figure 2) that use dependency patterns and clue words. In Figure 2, rectangles indicate a phrase and arrows show the dependency output using CaboCha, a Japanese dependency parser (Kudo and Matsumoto, 2002). The restrictions of dependency patterns were decided after investigating which Japanese particles were suitable for triplet extraction and which were causing noise. We also examined the percentage of particles in the set of 181 sentences containing triplets, and discovered that between the target and effect phrases, particles that indicate subject ("*ga*": 57.7%, "*ha*": 12.7%) and object ("*wo*": 11.0%) were most often used. For this reason, we used these three particles to restrict dependency

Table 1. Examples of sentences including the drug effect triplet

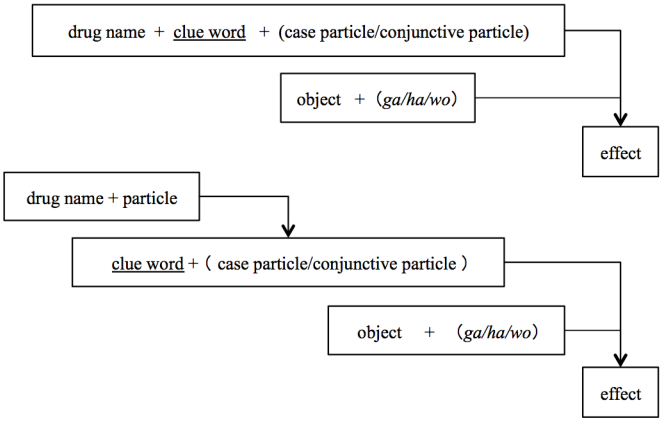
Sentence	Triplet
Sorekara puredonin o nomi hajimeta nodesuga, puredonin no fukusayō de hone ga yowaku nari (Then I started drinking Prednisone, but my bones were weakened by the side effects of Prednisone)	*puredonin, hone, yowaku* (Prednisone, my bones, weak)
Tabun jipurekisa de atama ga boke-gimina nda to omoimasu (Maybe, I (lit. my head) am going senile due to Zyprexa)	*jipurekisa, atama, boke-gimi* (Zyprexa, my head, senile)
Genzai no hossa wa, arebiachin no chūnyū jikan zengo (6-ji, 14-ji, 22-ji) ni keiren ga shutsugen shite iru (In the current attack, convulsions appeared around the time of injecting Aleviatin (6:00, 14:00, 22:00))	*arebiachin, keiren, shutsugen* (Aleviatin, convulsions, appear)

Table 2. Examples of clue words

fukusayō (side effect)	eikyō (effect)	okage (thanks to)
sei (because of)	ato (after)	tōyo (dose)
shiyō (usage)	fukuyō (take (medicine))	kaeru (change)
nomu (drink)	tsukau (use)	chūnyū (injection)

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Figure 2. Retrieval restrictions of proposed dependency patterns



patterns. In the preliminary research (Kitajima et al., 2013), the precision increased by 9.7 points when setting top three frequently used particles to restrict dependency patterns from using top six particles. The second heuristic for such restriction was based on causative expressions such as case and conjunctive particles. Case particles (e.g. "*de*", "*kara*", "*yori*") have a nuance of "continuance" or "process" of actions and states. Conjunctive particles indicate "cause" or "motive" for process. When these particles are constrained, the precision increased by 11.3 points (Kitajima et al., 2013).

3.4. Converting the Phrase

After detecting the dependency chunks containing a triplet element, our system converts the chunks into a simplified form. For example, it converts the words to dictionary form and omits particles. This is how our system extracts the dictionary form of the first word in the chunk as a triplet element. In addition, our system combines consecutive nouns such as "right" and "elbow". If a genitive particle precedes an object, it becomes a part of this object. Also, our system makes negative expressions from stemmed verb and the auxiliary "not". In the preliminary research, we discovered that about 7.6% out of 340 triplets contain the auxiliary "not", therefore, these triplets requires a conversion to stemmed verb and the auxiliary "not" in order to extract drug effect expressions correctly. In this study, our system does not consider "… *masen* (not)", negative clue words (e.g. "*kiru* (stop)", "*genryo* (reduce)") or contradictory conjunctions (e.g. "but", "however") as negative words. In future studies, we will deal with such words.

4. EVALUATION EXPERIMENT

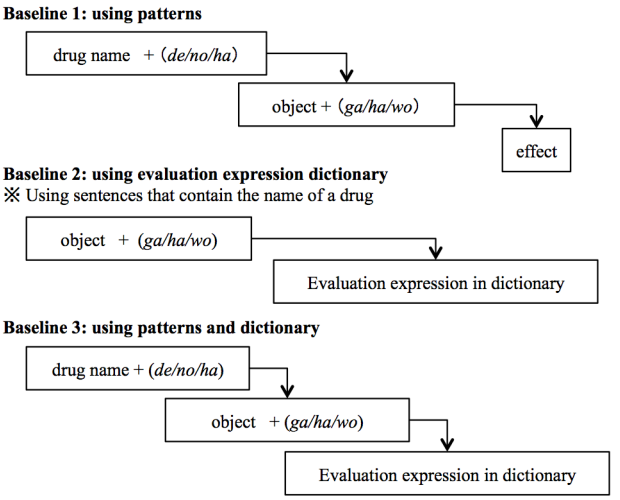
The purpose of our experiment was to demonstrate the extraction performance of the algorithm on blog data. The first author conducted judgments on the appropriateness of the triplets extracted by our system and baseline systems for information on drug effects. A detailed explanation of the baseline systems is described later in this section. We used precision, recall, and F- measure to evaluate. These can be calculated using the following three equations.

$$Precision = \frac{correct}{correct + mistake} \quad (1)$$

$$Recall = \frac{correct}{correct\,triplets\,by\,manual\,extraction} \quad (2)$$

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Figure 3. Baseline systems



$$F - measure = \frac{precision * recall * 2}{precision + recall} \quad (3)$$

4.1. Data

Our system extracts from 2,369 sentences containing drug names among 2,000 snippets which were collected by "TOBYO-*jiten*" ("dictionary"). TOBYO-jiten is a tool in TOBYO that searches blog articles using medical keywords. The 340 correct triplets of information on effects of drugs were obtained by manual extraction.

4.2. Baseline Systems

Shown in Figure 3 are three extraction methods used as baselines for comparison. Baseline system 1 extracts when the parsing results match the pattern "drug name (*de/no/ha*) → object" and "object (*ga/ha/wo*) → effect". In baseline systems 2 and 3, the extraction methods use words in EVALDIC_ver1.0.1 (Kobayashi et al., 2004), a general Japanese dictionary of evaluation expressions, as the evaluation factors. The dictionary contains about 5,234 words. In baseline system 2, the target elements are extracted from the phrase depending on the phrase that contains the evaluation factor (the word in EVALDIC_ver1.0.1) by "*ga, ha, wo*", from sentences that contain the name of the drug. Moreover, baseline system 3 extracts when a dependency relation using the particles "*de, no, ha*" is formed between the phrase that contains the drug name and the phrase that contains the target element.

4.3. Results and Discussion

The results of the evaluation are displayed in Table 3. Our proposed system achieved the highest precision, but the recall was very low. Other methods scored higher F-measures, and baseline system 2, which loosens dependency restrictions, achieved the highest recall. For patients' survival blogs, it was confirmed that extraction using clue words and particle restrictions is effective for achieving higher precision. In addition, from the results of the baseline systems 2 and 3, we confirmed that the number of extractions is dependent on the limitation of the relationship between the drug name and other elements. Table 4 shows respective output examples. The correct output example in Table 4 is interesting because the effect is an

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Table 3. Evaluation results

	Outputs	Correct outputs	Precision [%]	Recall [%]	F-Measure
Proposed system using clue words and retrieval restrictions	47	24	51.1	7.1	0.126
Baseline 1: using patterns	105	11	10.5	3.2	0.049
Baseline 2: using evaluation expression dictionary	866	97	11.2	28.5	0.161
Baseline 3: using patterns and dictionary	49	7	14.3	2.1	0.021

onomatopoeia, which is non-technical and is not included in the evaluation expression dictionary. This is not documented in drug descriptions, so such information cannot be collected from usual medical texts.

There are two types of errors in our system: misjudgments of the position of the phrase including the element, and unsuitable conversion of the element of the triplet.

The reasons why our system misjudged the position of phrases are parsing errors, which are attributed to incomplete sentences contained in snippets, and unmatched patterns of extraction due to missing postpositional particles because the blogs are written in spoken language. Our system could not extract triplets when the subject and particles were omitted because blogs are written in spoken language. Therefore, it is necessary to consider a new method which does not depend on particles or extracts after supplementing particles. Dealing with such omissions will be included in our future work. In addition, incomplete sentences caused by wrongly cut snippets caused parsing errors which we plan to reduce by using the full texts of articles, not only snippets. Moreover, some sentences could not be extracted from, owing to the use of unregistered clue words. In future studies, we will increase the number of clue words. Additionally, due to nonexistence of clue words and mismatch of the extraction patterns, some correct triplets extracted by the baseline systems 1 and 3 could not be extracted by our system. Thus, it can be considered that integration of the methods of the baseline systems into our system would be effective to increase recall. Additionally, in future our system needs to recognize that "by taking a medicine called Baclofen" is essentially the same as "by taking Baclofen", and to consider the combination of drugs, in order to achieve correct parsing.

Table 4. Examples of our system outputs

Sentence	Output	Correct
Saiwai ni mo chiryō ga susumu ni tsurete jiriki de jinzō ga ugoki de rete, sonomama shinkyū ga dekimashita kōchō sensei arigatō demo suteroido no fukusayō de kao wa panpan (Fortunately, with the progression of my treatment, my kidneys started to work again, and I was promoted. Thanks, principal. But my face was swollen due to a side effect of the steroids)	*(suteroido, kao, panpan)* (steroids , face , swollen)	*(suteroido, kao, panpan)* (steroids , face , swollen)
Watashi ni totte takisotēru no datsumō-ritsu wa kekkō takai (For me, the rate of hair loss with Taxotere is quite high)	*(takisotēru, (empty) , (empty))* (Taxotere , (empty) , (empty))	*(takisotēru, datsumō-ritsu, takai)* (Taxotere , the rate of hair loss , high)

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In order to reduce output errors, it is necessary to judge the appropriateness of the elements for the triplets of drug effect information. We often find that a triplet is wrong because the extracted object element was inadequate (e.g. error target: "*kakuritsu* (probability)", correct target: "*kami ga nukeru kakuritsu* (probability of one's hair falling out)"; error target: "1*nen-ijou* (more than 1 year)", which is simply a period of time). When we describe the effects of drugs, we use various evaluation words, but the objects in which the change or effect of the drug occurred are limited to some extent. Therefore, to solve these issues we believe it will be effective to create a dictionary which contains words used as the object when we explain drug effects, and using machine learning to convert the phrase into an appropriate form and range, as a triplet of drug effect information. In addition, our system does not consider "…masen (not)", negative clue words (e.g. "*kiru* (stop)", "*genryo* (reduce)") as negative words. In future studies, we will deal with such words.

5. IMPROVED RECALL WITH INCORPORATED SYSTEM

We designed a new system that extracts using baseline system 2. While our method from the previous proposed system cannot extract triplets when there is no clue word in a sentence, our newly proposed system extracts triplets using the dependency pattern of baseline system 2, which has loosen dependency restrictions and scores the highest recall.

5.1. Evaluation Results and Discussion

We performed an evaluation experiment for the dataset used in the first evaluation experiment. Table 5 shows the evaluation results. By combining the proposed method 1 and baseline system 2, the recall improved by 26.1 points, and achieved the highest F-measure. Output examples of the integrated system are shown in Table 6. The previous system could not extract the triplet that is shown at the top of Table 6 because dependency patterns could not be applied. However, the integrated system can extract more data due to combining clue words, particle restrictions and loose dependency restrictions. Another example shown in Table 6 is incorrect output caused by wrongly cut snippets and by the unsuitability of extracted elements. We plan to reduce the parsing errors by using the full texts of articles, not only snippets. Additionally, we plan to judge the appropriateness of the element extracted as a drug effect. In short, pattern matching with clue words and particle restrictions was effective for improving precision, and the combination of baseline system 2 and the proposed method 1 was effective for improving recall. However, a decrease in precision obviously accompanied the increase in recall. In future we need to keep the precision from decreasing, for example by using some kind of knowledge base.

6. IMPROVED PRECISION WITH CREATED DICTIONARIES

When patients describe taking medicine there are various evaluation expressions, however, it can be assumed that the number of objects of change or effect is somewhat limited. Therefore, we constructed two target word dictionaries which contain words used as a target that indicate a state or a body part, collected from existing documents written about medicine. The dictionaries basically register nouns or pairs of a prefix and a noun. They also register consecutive nouns as a word. If a word contains a number, we consider the word to be unsuitable for our target dictionaries (e.g. "1hour"). We utilized MeCab (Kudo, 2005) as a Japanese morphological analyzer. Subsequently, we judged the appropriateness of the elements extracted as drug effects using "110 dictionary" which is a dictionary we made from the website "*okusuri* (medicine) 110-ban (the Japanese emergency service telephone number) (Pharma Friend, 2013)" and "MPi dictionary" which is a dictionary we created from medical package inserts. We judge whether a word is

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Table 5. Integrated system result

	Outputs	Correct outputs	Precision [%]	Recall [%]	F-Measure
Proposed system using clue words and retrieval restrictions	47	24	51.1	7.1	0.124
Baseline2: using evaluation expression dictionary	866	97	11.2	28.5	0.161
Integrated system (proposed system + Baseline 2)	890	113	12.7	33.2	0.184

relevant for a triplet if part of the target element is included in the dictionaries.

6.1. 110 Dictionary

The NGO-operated website "*okusuri* 110-ban (Pharma Friend, 2013)" provides information about medicines, their usage, misusage, dosing, and so on. It is a site for patients and their families, written in casual language and easy to understand. We extracted 12,170 consecutive nouns from most of the website's sections, for example describing a drug's basic information, its efficacy, its side effects, and so on. Table 7 shows a selection of these words.

6.2. MPi Dictionary

We created the MPi dictionary by utilizing words from medicine package inserts of prescription drugs and OTC (Over-the-Counter) drugs. This information can be collected from the Pharmaceuticals and Medical Devices Information website (Pharmaceuticals and Medical Devices Agency, 2013), managed by the Pharmaceutical and Medical Devices Agency. Medicine package inserts are created by manufacturers and selling distributors of pharmaceuticals and use various technical terms. As the purpose of using this dictionary is to filter triplets extracted from blogs collected by "TOBYO-jiten", we

Table 6. Output examples of integrated system

Sentence	Output	Correct
Rifurekkusu fukuyō-go no muzumuzu-kan ga nakunatta koto o tsutaeta ato no sensei karata ((…) from the doctor after I said that the itching feeling has gone after taking Reflex)	*(Rifurekkusu, muzumuzu-kan, nakunaru)* (Reflex , itching feeling , has gone)	*(Rifurekkusu, muzumuzu-kan, nakunaru)* (Reflex , itching feeling , has gone)
Yokatta zutsū ni tsuite wa, puredonin o tōyo shi hajimete kyokudo ni shiryoku ga sagatte kite ite kontakuto (It's good. Re: my headache, my sight has got drastically worse since I started taking Prednisone)	*(puredonin, shiryoku, sagaru)* (Prednisone , sight , worse)	*(puredonin, shiryoku, sagaru)* (Prednisone , sight , worse)
Ima made kōjōsen horumon ga takusandete itaga, ikura kanshoku bōin bōshoku shite mo taijū ga fuenai hazuda! ! Merukazōru no okage de byōki jitai wa yoku natte kitamonono watashi no karada ga dondo (Until now, I had a lot of thyroid hormone, and my weight should not increase after drinking and eating too much!! My disease is getting better thanks to Methimazole but my body rapidl…)	*(Merukazōru, watashi no karada, dondo)* (Methimazole , my body , rapidl)	empty
(Hidarigawa-me ni kizu ga tsuite ita sei de tengan kusuri o dashite moraimashita (hiarein) jikai no yoyaku wa 3-shūkan-godesu tantō no sense wa 9 tsuki 30 hidzuke de i (I got eyedrops (Hyalein) because my left eye was damaged, my next appointment is 3 weeks later, on September 30)	*(hiarein, yoyaku, i)* (Hyalein, appointment,)	empty

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Table 7. Examples from 110 dictionary

Nemuri (sleep)	*Keiren hossa* (convulsive attack)	*Shibire* (numbness)	*Fukigen* (bad mood)	*Sōutsubyō* (depressive illness)	*Teashi* (hands and feet)	*Henzutsū* (migraine)
me (eye)	*Fukusayō* (side effect)	*Zutsū* (headache)	*Nodo* (throat)	*Kibun* (feeling)	*Hakike* (nausea)	*Shōjō* (condition)
Kayumi (itch)	*Nemuke* (sleepiness)	*Shūchū-ryoku* (concentration)	*Hatsunetsu* (fever)	*Hosshin* (rash)	*Nō* (brain)	*Monowasure* (failure of memory)

constructed "MPi dictionary" from descriptions of drug efficacy and side effects, which are a section of 296 medicine package inserts. The 296 drug names were used as keywords to collect snippets with "TOBYO-jiten". The MPi dictionary covers 5,380 words, part of which is shown in Table 8.

6.3. Evaluation Results and Discussion

We performed an evaluation experiment for the dataset used in the first evaluation experiment. Table 9 and Table 10 shows the result.

When the words in the dictionaries were used for filtering objects in our proposed systems, the precision increased by 8.9 points. For example, the system accepted "sleep in the morning" as a correct object, but it filtered out "more than one year and a half" because it was not in our new dictionary. This confirmed that filtering using the target dictionary was effective. We can infer that the 110-dictionary is more appropriate for blogs written in casual language. The MPi-dictionary was very restrictive, because it registers fewer words and a large number of technical terms. The dictionaries include some noise such as "*koto* (thing)", "*no* (of)", and the particle "*wo*" due to a lack of consideration of word semantics. Data cleaning of our dictionaries is required in order to increase the precision when utilizing the dictionaries.

7. ANNOTATION

As there is a risk that evaluation would be biased if conducted by a single evaluator, we created a manually corrected version of the data using human annotators, and performed an evaluation experiment against that data. We created this corrected version based on data which was annotated by five college students in their early twenties. The annotators were two science graduate students and three science undergraduate students, three men and two women, all native speakers of Japanese. They

Table 8. Examples from MPi dictionary

Suimin (slumber)	*Shōtenhossa* (focal seizure)	*Shibire* (numbness)	*Fukigen* (bad mood)	*Koto* (thing)	*Hana* (nose)	*Kinryoku teika* (muscle weakness)
CPK (CPK)	*Fukusayō* (side effect)	*Zutsū* (headache)	*Hassei shōgai* (dysphonia)	*Shokku* (shock)	*Toketsu* (hematemesis)	*Shōjō* (condition)
infusion reaction (infusion reaction)	*Keimin* (drowsiness)	*benpi* (constipation)	*shisei* (position)	*Hosshin* (rash)	*Nō* (brain)	*kenbō* (forgetfulness)

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Table 9. Dictionary filtering results

	Outputs	Correct outputs	Precision [%]	Recall [%]	F-measure
110 dictionary filter					
Proposed system using clue words and retrieval restrictions	35	21	60.0	6.2	0.112
Integrated system (proposed system + Baseline 2)	524	92	17.6	27.1	0.213
Baseline 1: using patterns	67	8	11.9	2.4	0.039
Baseline 2: using evaluation expression dictionary	494	72	14.6	21.2	0.173
Baseline 3: using patterns and dictionary	30	4	13.3	1.2	0.022
MPi dictionary filter					
Proposed system using clue words and retrieval restrictions	20	9	45.0	2.7	0.050
Integrated system (proposed system + Baseline 2)	237	44	18.6	12.9	0.153
Baseline 1: using patterns	28	5	17.9	1.5	0.027
Baseline 2: using evaluation expression dictionary	219	35	16.0	10.3	0.125
Baseline 3: using patterns and dictionary	11	2	18.2	0.6	0.011

annotated the positions of triplets that express drug effects within 2,000 sentences containing drug names among snippets which were newly collected using "TOBYO-jiten". We obtained 258 correct triplets which were identified by two or more annotators in agreement, and considered these to be the "correct triplet elements". The range of the correct triplet elements tagged by annotators are not exact matches, but are semantically equivalent.

7.1. Annotation Results

The κ-value calculated from the annotation data by our five annotators was 0.41, which was very low. In order to analyze the reason for few exact matches, the first author, who has an understanding of the purpose of our system, manually tagged triplets in 200 sentences among 2,000 sentences to compare annotation with that of the five annotators. As

Table 10. Output examples filtered by dictionaries

Sentence	Output	Appropriate ?		Correct
		110 dictionary	MPi dictionary	
Suimin wa, nerubon jō kara yūrojin jō ni kaete kara asa no nemuri ga asaku natta yō (My sleep in the morning seems to be lighter since I have changed from Nervon to Eurodin)	*(yūrojin, asa no nemuri, asai)* (Eurodin, sleep in the morning, light)	Yes	No	Yes
Pegashisu tandoku chiryō o shūryō shite 1-nen han ijō o keika shita (More than one year and a half has passed since the Pegasys monotherapy finished)	*(pegashisu, 1-nen han ijō, keika)* (Pegasys , more than one year and a half, passed)	No	No	No
Maisurī o nomu to tsuyoi kentaikan ga deru node, kinō wa nomazu ni shūshin shimashita (I went to bed without taking Myslee because I feel a strong fatigue when I take it.)	*(Maisurī, kentaikan, deru)* (Myslee, fatigue, feel)	Yes	Yes	Yes

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Table 11. Annotation results for 200 sentences

Annotators in agreement	Extracted triplets	Comparison with first author		
		Concerted triplets	Precision [%]	Recall [%]
Over three	12	12	100.0	37.5
Over two	22	20	90.9	62.5
Over one	53	30	56.6	93.8

a result, for these 200 sentences, the number of triplets tagged by the first author was 32, while the mean value of triplets identified by the five annotators was 17. The primary cause for this appears to be that some annotators did not fully understand the purpose of our system. We wanted to extract medicinal effects that appeared *and* effects that did not appear, however, some annotators tagged only effects that appeared. For example, none of annotators tagged the triplet (Reflex, itching feeling, has gone) for the sentence "thanks to Reflex, the itching feeling has gone". Considering our system's purpose, the triplet (Reflex, itching feeling, has gone) is suitable to be tagged as correct data. Moreover, another major factor for the low κ-value was that the annotators were using different ranges of triplets. For example, in the sentence "For me, the rate of hair loss with Taxotere is quite high", one annotator tagged "high" as the effect, but another annotator tagged "quite high" as the effect. In this way, one person annotated only the adjective as an effect, whereas another used the pair of adverb and adjective. When we regard the triplets which have different ranges as equal if they are semantically equivalent, the number of different triplets which collected by five annotators from 200 sentences was 53.

Below, we cite some examples in which the triplets tagged by annotators and the triplets identified by the first author do not match. The annotators were confused when the blogger used incorrect expressions and when the sentence was incomplete because it was a piece of a snippet, for example "*Teido wa mashi* (the level is good)" (here, the correct notation is "*Aruteido wa mashi* (relatively good)"). Judging the intentions of the blogger is often difficult for the system; in fact, it is sometimes even difficult for humans. For example, we cannot comprehend that the writer of "This result is the lowest since I started Prednisone except when I caught a cold" actually meant "I caught a cold as the result of starting Prednisone". In addition, it was shown that Voltaren has the effect of lowering body temperature by the snippet "If I get a high temperature, this is a hospital, so surely someone will give me an appropriate treatment such as Voltaren"; however, some annotators annotated the triplet (Voltaren, temperature, high). It is difficult for the system to make such judgments, as the system needs to conduct reasoning using not only superficial information but also common sense in order to extract the effects of Voltaren in this sentence. Although we plan to make a system capable of inference in further studies, this issue is not included in the scope of the present study.

Table 11 shows the result of the concordance between the first author and annotators. We regarded 258 triplets as correct due to the high concordance between the first author and two annotators. For the future work, it is necessary to carefully demonstrate the aim of our research and the annotation methods to participants in order to create an appropriate annotation data.

7.2. Evaluation Results and Discussion

We performed an evaluation experiment using the data created by the annotation process (hereafter, the "annotation data"). We used precision, recall, and F-measure to evaluate. Precision and F-measure can be calculated using

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Table 12. Evaluation results for annotation data

	Outputs	Correct outputs	Precision [%]	Recall [%]	F-measure
Proposed system using clue words and retrieval restrictions	38	16	42.11	6.20	0.11
Integrated system (proposed system + Baseline 2)	681	80	11.75	31.01	0.17
Baseline 1: using patterns	143	2	1.40	0.78	0.01
Baseline 2: using evaluation expression dictionary	662	64	9.67	24.81	0.14
Baseline 3: using patterns and dictionary	35	1	2.86	0.39	0.01
110 dictionary filter					
Proposed system using clue words and retrieval restrictions	29	15	5.81	0.10	5.81
Integrated system (proposed system + Baseline 2)	398	65	25.19	0.20	25.19
Baseline 1: using patterns	110	2	0.78	0.01	0.78
Baseline 2: using evaluation expression dictionary	379	50	19.38	0.16	19.38
Baseline 3: using patterns and dictionary	28	1	0.39	0.01	0.39
MPi dictionary filter					
Proposed system using clue words and retrieval restrictions	19	11	4.26	0.08	4.26
Integrated system (proposed system + Baseline 2)	181	38	14.73	0.17	14.73
Baseline 1: using patterns	49	0	0.00		0.00
Baseline 2: using evaluation expression dictionary	166	27	10.47	0.13	10.47
Baseline 3: using patterns and dictionary	11	0	0.00		0.00

the previous equations (1), (3), and Recall can be calculated using the following equation. The "correct triplets determined by annotation" in the equation is the aforementioned 258 triplets.

$$Recall = \frac{correct}{correct\,triplets\,determined\,by\,annotation} \quad (4)$$

Table 12 shows the results of the evaluation for the annotation data. Since the annotation data was not fully suitable for our purpose, the evaluation value was generally low. However, it was confirmed that our system's extraction patterns are effective for illness survival blogs, the integrated system is useful for increasing the recall, and the target dictionaries are efficacious for improving the precision. From the results of annotation, it was demonstrated that the information regarded as important depends on the individual annotator. We can consider that extraction of phrases may be more effective than extraction of triplets for providing medical information.

8. CONCLUSION AND FUTURE WORKS

We have presented an approach to extract information on the effects of drugs from snippets of illness survival blogs. What is different from other approaches is that our system extracts triplets using dependency patterns and clue words so that we can extract both positive and negative effects of drug usage from texts which contain many kinds of topic and are written in Japanese casual language. We confirmed that our system extracts by 39.9 points higher precision than existing methods. We incorporated the method of the baseline system 2 into our system and verified the recall increased by 26.1 points. Furthermore, we filtered the target elements using dictionaries which we constructed from existing medical documents and discovered the precision improving by 8.9 points.

In future works, we will increase the number of the clue words in order to reduce parsing errors and match the extraction pattern. We also need to restructure the annotation data because the information regarded as important depends on the individual. Additionally, we will consider developing a system that extracts information on effects of drugs using phrases instead of triplets, as some of the information expressed by triplets is unsatisfactory. We will also change the target data from snippets to full blog articles. After extraction, the next aim of our system is determining the modality of the information (e.g. hearsay, experience, supposition). The purpose of our research is to make a system that allows the sharing of narrative information between patients easily, and helps patients to collect medical information.

Additionally, health data exchange online is growing not only in the form of text but also as output for sensor devices. We believe that the more people share such information, the greater chance there is that we will be able to discover patterns that doctors cannot yet find.

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REFERENCES

Akiko, O., Miki, S., & Ogawa, M. (2007). Anonymity enhances the appearance of word-of-mouth on the Internet: From the survey on "MICRODIET" users. [in Japanese]. *The Journal of the Infosocionomics Society*, *2*, 114–127.

Aramaki, E., Maskawa, S., & Morita, M. (2011). Twitter catches the flu: Detecting influenza epidemics using twitter. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 1568–1576).

Aramaki, E., Miura, Y., Tonoike, M., Ohkuma, T., Mashuichi, H., Waki, K., & Ohe, K. (2010). Extraction of adverse drug effects from clinical records. *Studies in Health Technology and Informatics*, 739–743. PMID:20841784

Cortes, C., & Vapnik, V. (1995). Support vector networks. *Machine Learning*, *20*, 273–297. doi:10.1007/BF00994018

Fung, K. W., Jao, C. S., & Demner-Fushman, D. (2013). Extracting drug indication information from structured product labels using natural language processing. *Journal of the American Medical Informatics Association*, *20*, 482–488. doi:10.1136/amiajnl-2012-001291 PMID:23475786

Green, D. M., & Swets, J. A. (1966). *Signal detection theory and psychophysics* (Vol. 1). New York, NY: Wiley.

Hesse, B. W., Nelson, D. E., & Kreps, G. L. et al. (2005). Trust and sources of health information: The impact of the internet and its implications for health care providers: Findings from the First Health Information National Trends Survey. *Archives of Internal Medicine*, *165*(22), 2618–2624. doi:10.1001/archinte.165.22.2618 PMID:16344419

Kaiser, C., & Bodendorf, F. (2012). Mining patient experiences on web 2.0 - A case study in the pharmaceutical industry. In *Proceedings of the SRII Global Conference 2012* (pp. 139-145).

Kitajima, S., Araki, K., & Rzepka, R. (2012). An extraction method of medication usage information using clue words from illness survival blogs. In *Proceedings of the Fuzzy System Symposium* (Vol. 28, pp. 251-256) (in Japanese).

Kitajima, S., Rzepka, R., & Araki, K. (2013). Extraction of drug information using clue words from Japanese blogs. In *Proceedings of the 14th Conference of the Pacific Association For Computational Linguistics 2013* (Paper Sep4-8), Tokyo, Japan.

Kitajima, S., Rzepka, R., & Araki, K. (2013). *Improvement of drug effects extraction system from patient's blogs using collection of clinical terms*. Special Interest Group on Language Sense processing Engineering (SIG-LSE), SIG-LSE-B301-4, 19-23 (in Japanese).

Kitajima, S., Rzepka, R., & Araki, K. (2013). Performance improvement of drug effects extraction system from Japanese blogs. In *Proceedings of 2013 IEEE Seventh International Conference on Semantic Computing*, Irvine, CA (pp. 383-386).

Kobayashi, N., Inui, K., Matsumoto, Y., Tateishi, K., & Fukushima, T. (2004). Collecting evaluative expressions for opinion extraction. In *Proceedings of IJCNLP* (pp. 584–589).

Kokubu, H. (2008). Difficulties experienced by breast cancer patients in selecting their initial treatment. *Japanese Society of Cancer Nursing, 22*(2), 14-22 (in Japanese). Initiative Inc. (n.d). *TOBYO: Nihon no Tobyo ki* 40000. Retrieved December 9, 2013, from http://www.tobyo.jp/

Kudo, T. (2005). *Mecab: Yet another part-of-speech and morphological analyzer*. Retrieved from http://mecab.sourceforge.net/

Kudo, T., & Matsumoto, Y. (2002). Japanese dependency analysis using cascaded chunking. In *Proceedings of the 6th Conference on Natural Language Learning 2002 (CoNLL 2002)* (pp. 63–69).

Maeda, Y., Kokufu, H., & Fujii, T. (2009). The influence of fellow patients for patients with breast cancer during treatment: Survey on the self-help group. [Japanese Society of Cancer Nursing]. *Gan-kango*, *14*(6), 711–716.

Matsumoto, A., Aragaki, M., Zheng, Y. F., Inokuchi, T., Sakai, K., & Shiota, S. et al. (2005). How Japanese breast cancer survivors perceive the internet as a health information resource: An internet survey with a mailing list participants. *Japan Journal of Medical Informatics*, *25*(3), 151–159.

Mikami, T., Komada, Y., Noguchi, T., Sugano, A., & Yamana, H. (2013). Iyakuhin fukusayō jōhō o mochiita fukusayō kensaku shisutemu no teian (Proposal for a Side Effects Retrieval System Using Pharmaceutical Side Effect Information). [in Japanese]. *Information Processing Society of Japan*, *IFAT-111*(11), 1–6.

Ministry of Health, Labour, and Welfare of Japan (Research on Patient Survey (Total Number of Patients). (2011). Retrieved from http://www.mhlw.go.jp/toukei_hakusho/toukei/

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NPO Narrative of Disease and Health DIPex-Japan. (n.d). Retrieved December 9, 2013, from http://www.dipex-j.org/

O'Connor, A. M. (2002). An evidence-based approach to managing women's decisional conflict. *Journal of Obstetric, Gynecologic, and Neonatal Nursing*, *31*(5), 570–581. doi:10.1111/j.1552-6909.2002.tb00083.x PMID:12353737

Pharma Friend (K. Kobayshi). (n.d.). *Okusuri 110-ban*. Retrieved December 9, 2013, from http://www.jah.ne.jp/~kako/frame_dwm_search.html

Pharmaceuticals and Medical Devices Agency. (n.d). Retrieved December 9, 2013, from http://www.info.pmda.go.jp/index.html

Setoyama, Y., & Nakayama, K. (2011). Literature review for information needs, information resources, and difficulties of information use among patients with breast cancer in Japan. [Health Care and Society]. *Iryo To Shakai*, *21*(3), 325–336. doi:10.4091/iken.21.325

Shinohara, E., Hattori, K., Miura, Y., Tonoike, M., Ohkuma, T., & Mashuichi, H. … Ohe, K. (2011). Koubun Patahn ni Motozuku Yakuzai Fukusayou Zyouhou no Zidou Chuushutsu (Automatic Extraction of Drug Side Effects Using Syntactic Patterns). In *Proceedings of the 31st Joint Conference on Medical Informatics* (pp. 521-524) (in Japanese).

Sugiki, K., & Matsubara, S. (2008). A product retrieval system robust to subjective queries. *International Journal of Product Lifecycle Management*, *3*(2-3), 151–164. doi:10.1504/IJPLM.2008.021439

Taniguchi, A. (2004). Ikki teki nyuubou saiken-jutsu o uketa nyūgan josei no omoi to kango shien (Nursing Support and Mindset of Women after Undergoing Breast Reconstruction). [in Japanese]. *Journal of Japan Academy of Nursing Education*, *45*(8), 660–663.

Tsuchida, M., Mizuguchi, H., & Kusui, D. (2009). Ranking method of object-attribute-evaluation three-tuples for opinion retrieval. In *Proceedings of the New Frontiers in Artificial Intelligence: JSAI 2008 Conference and Workshops* (vol. 5447, pp. 87-98).

Tsuchiya, M., & Horn, S. (2009). An exploration of unmet information needs among breast cancer patients in Japan: A qualitative study. *Journal of Japanese Society of Cancer Nursing*, *18*(2), 149–155. PMID:19267730

Walsh, M. C., Trentham-Dietz, A., Schroepfer, T. A., Reding, D. J., Campbell, B., & Foote, M. L. et al. (2010). Cancer information sources used by patients to inform and influence treatment decisions. *Journal of Health Communication*, *15*(4), 445–463. doi:10.1080/10810731003753109 PMID:20574881

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